

Mock Exam

CS 101 Algorithmic Problem Solving L2

Fall 2023

Name(s): _____

HU ID (*e.g., xy01042*): _____

Exam Instructions

- Exam Duration: 2 hours and 30 minutes
- Number of Questions: This exam contains 4 coding questions and 1 debugging question.
- Technical Requirements: You are not allowed to use code editors, compilers, or online resources during this exam. Only electronic calculators are permitted during this exam.
- Communication: During the exam, do not communicate with anyone except the instructor or exam proctors if you have questions about the exam.
- Honesty: Academic honesty is expected. Do not engage in any form of cheating, plagiarism, or unauthorized collaboration.
- Answer Format: Write your answers clearly in the provided spaces.
- Time Management: Monitor the time carefully and pace yourself. Do not spend too much time on any one question or section.
- Exam Completion: Before submitting, take a moment to review your answers to ensure you have completed all questions and that your responses are clear and accurate.

Good Luck!

1. Friend's share

Areeba is buying some packs of gum that cost \$1.50 each. She wants to share her gum with her friends. She asks her friends to pay her for their share. Including Areeba, how much does each person spend on gum?

Write a function `friends_share()`, that takes two parameters, `no_of_packs`, and `no_of_friends`, and prints each person's share.

Constraints

- $1 \leq \text{no_of_packs} \leq 99$
- $1 \leq \text{no_of_friends} \leq 10$

Interaction

The input consists of two space-separated numbers representing `no_of_packs`, and `no_of_friends` respectively. The output should contain a string that states: "Each friend spent X dollars and Y cents." where X and Y are the actual amount spent in dollars and cents respectively.

Sample

Input	Output
5 2	Each friend spends 2 dollars and 50 cents
10 5	Each friend spends 2 dollars and 50 cents

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
6 7	✓ 1 dollar 12 cents
4 2	2 dollars 0 cents
99 10	13 dollars 15 cents

Problem Identification

Briefly explain the problem you identified in the above question that led you to your solution.

input: `no_of_packs`, `n_of_friends` . Output: cost of the packs divided by the no of friends+1.

Pseudocode

```
def friends_share(no_of_packs,no_of_friends):
    cost=(no_of_packs*1.5)/(no_of_friends+1)
    cost_int=(no_of_packs*1.5)//(no_of_friends+1)
    cents=cost-cost_int
    print("Each friend spends",int(cost_int), "dollars and ",int(cents*100),"cents")

friends_share(5,2)
```

Dry Run

Using any **one** input provided in the **Exercise section** above, dry run your pseudocode in the space below.

2. Time to Chill

A specific month has 30 days, numbered from 1 to 30. Day 1 is a Monday and the usual A 7-day week is followed (so day 2 is Tuesday, day 3 is Wednesday, and so on). Every Saturday and Sunday is a holiday. There are N public holidays.

Note that it is possible for a public holiday to occur on a Saturday or Sunday.

Samad wants to know the total number of holidays he would get that month. Given the dates of public holidays, write a function *Holidays* that returns the total number of holidays. The function takes N as an argument and takes the dates as input N times.

Constraints

- $1 \leq N \leq 10$

Interaction

The input comprises a single line containing an integer denoting the value of N . It is then followed by N lines which each state the public holiday date. The output must contain a single number denoting the total holidays.

Sample

Input	Output
2	9
5	
7	
3	10
23	1
1	
7	

In the first case, Days 6, 13, 20 and 27 are Saturdays, and days 7, 14, 21, 28 are Sundays. The public holidays fall on day 5 and day 7, but day 7 is already a Sunday. This gives us 9 holidays.

In the second case, Days 6, 13, 20 and 27 are Saturdays, and days 7, 14, 21, 28 are Sundays. The public holidays fall on day 1, day 6, and day 23. This gives us 10 holidays in total.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
1	8
13	
3	10
2	
27	
29	

Problem Identification

Briefly explain the problem you identified in the above question that led you to your solution.

Pseudocode

```
def Holidays(n):
    total_holidays=8
    for i in range(n):
        day=int(input())
        if day!=6 and day!=13 and day!=20 and day!=27 and day!=7 and day!=14 and day!=21 and
day!=28:
            total_holidays+=1
    return total_holidays
```

Dry Run

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

3. Triangle Number Sequence

A triangle number sequence is given as

1, 3, 6, 10, 15, 21, 28, 36, 45, ...

and can be derived geometrically by looking at the triangles in the figure below.

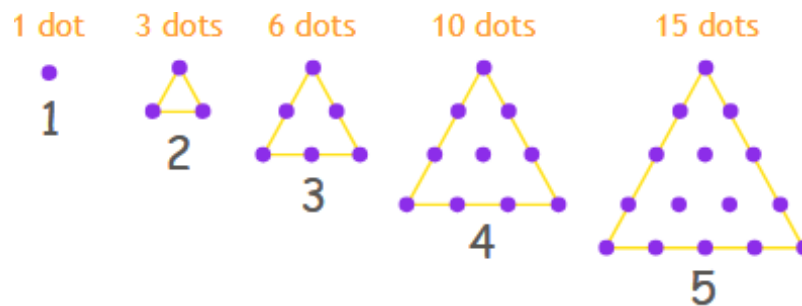


Figure 1: Triangular number Sequence

Write a function named `triangle_number` that takes a parameter `n` and returns the corresponding term in the above series.

Constraints

- is instance(`n`, int)
- $1 \leq N \leq 20$

Interaction

The input comprises a single line containing a single integer denoting the value of N .

The output must consist N lines, each containing the following pattern.

Sample

Input	Output
3	6
4	10

In the first case, $N = 3$. so you need to calculate the value of the Third term in the series which is calculated as the Value of the Second Term +3 i.e. $3+3=6$ dots.

In the second case, $N = 4$. so you need to calculate the value of the Fourth term in the series which is calculated as the value of the Third Term + 4 i.e. $6+4=10$

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
6	21
7	28

Pseudocode

```
def triangle_number(n):  
    count=0  
    for i in range(1,n+1):  
        count=count+i  
  
    return count
```

Dry Run

Using any **one** of the inputs provided in the **Exercise section** above, dry run your pseu-

docode in the space below.

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

4. Student Award

Habib University recognizes its students' efforts in the form of various awards that are given at the end of each academic year. Students are shortlisted for these awards based on some of the following criteria:

1. The student's academic performance has been above average i.e. CGPA is 3.5 or above
2. The student has been in leadership role in more than 2 students' clubs.
3. The student has been part of more than 3 community engagement projects.
4. The student bears good standing which means that there is no plagiarism or conduct case reported against him

On the basis of above criteria, the candidates for awards are selected. A student is a candidate for:

- Dean's award if conditions 1 and 4 are met.
- Leadership award if conditions 2 and 4 are met.
- Service award if condition 3, and 4, but not 2 are met.
- Yohsin award if all four conditions are met.

Challenge

Write a program that takes as input the following information about a student and prints a list of awards that he can compete for.

- CGPA
- Number of clubs led (leadCount)
- Number of community projects done (projects)
- Any plagiarism case against the student: (Yes/No) (plagiarism)
- Any report of misconduct: (Yes/No) (misconduct)

Additionally, if all the conditions are met except the last one, then the program gives this message. “Our code of conduct is definitive, it is not negotiable.”

Constraints

- $0 \leq CGPA \leq 4.0$
- 'leadcount' is int
- 'projects' is int
- 'plagiarism' is a string and should contain either 'Yes' or 'No'
- 'misconduct' is a string and should contain either 'Yes' or 'No'

Interaction

The input comprises a single line containing five space-separated items mentioned in the order above

The output must contain multiple lines stating the names of awards the student gets.

Note: If the Student doesn't qualify for any award nothing is printed.

Sample

Input	Output
3.5 2 5 No No	Deans Award
	Service Award
3.1 3 4 Yes No	

In the first case, the student's CGPA is 3.5 and there is no plagiarism or conduct case against him so he is a candidate for the Dean's Award.

Also, he is part of 5 community projects and no plagiarism or conduct case is against him, He has been in a leadership role for only 2 projects so he is also a candidate for the Service Award.

In the second case, only Conditions 2 and 3 are met which doesn't make him qualify for any Award.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
3.5 3 4 No No	
3.6 4 4 No Yes	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode

Dry Run

Using any **one** of the inputs provided in the **Exercise section** above, dry run your pseudocode in the space below.

5. Pattern

Write an iterative function named `pattern` to generate the pattern for a parameter, `n`.

Constraints

- $2 \leq N \leq 10$
- you are only allowed to use while loop to draw this pattern

Interaction

The input comprises a single integer which is the value of N

The output must show the pattern as explained in the sample section

Sample

Input	Output
4	* * * *
	* * *
	* *
	*
5	* * * * *
	* * * *
	* * *
	* *
	*

Proposed Solution

```
1 def Pattern(n):  
2     while n>1:  
3         i=0  
4         while i<=n:  
5             print("*",end="")  
6             i+=1  
7         print()  
8         n+=1  
9
```

Dry Run

Using one of inputs provided in the Sample section above, dry run the proposed code solution below.

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and the errors in each line.

Correct Solution

Rewrite the lines of code you mentioned above with their errors corrected.

Rough Work